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ABSTRACT

Objective: Hamstring injuries are common in field-based sports and reinjury rates are high. Recent evidence suggests hamstring injuries often occur during accelerative running, but investigations of hamstring mechanics have primarily considered constant-speed running. Thus, our objective was to compare hamstring lengths and velocities between accelerative running and constant-speed running. **Methods:** We recorded videos of 10 participants during 6 accelerative running trials and 6 constant-speed running trials. We used OpenCap to estimate body segment kinematics and a 3-dimensional musculoskeletal model to compute peak length and step-average lengthening velocity of the biceps femoris (long head) muscle-tendon unit. We compared running conditions using linear mixed models with running speed as the independent variable. **Results:** At running speeds below 75% of top speed, accelerative running resulted in greater peak lengths than constant-speed running. For example, the peak hamstring muscle-tendon length when a person accelerated from running at only 50% of top speed was equivalent to running at a constant 88% of top speed. Lengthening velocities were greater during accelerative running at all running speeds. Differences in hip flexion kinematics drove the greater peak lengths and lengthening velocities observed in accelerative running. **Conclusions:** Hamstrings are subjected to longer lengths and faster lengthening velocities in accelerative running than in constant-speed running. This provides a potential biomechanical perspective towards understanding the occurrence of hamstring injuries during acceleration. Our results suggest coaches and sports medicine staff should consider the accelerative nature of running in addition to running speed to quantify exposure to high-risk circumstances with long lengths and fast lengthening velocities of the hamstrings.

Key Words: BIOMECHANICS, KINEMATICS, MARKERLESS MOTION CAPTURE, MUSCLE INJURY, MUSCLE STRAIN, MUSCULOSKELETAL MODEL

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INTRODUCTION

Hamstring strain injuries affect many athletes (1–5) and comprise 10% of all injuries in field-based sports (6). Despite a plethora of research aimed at reducing injury rates, they have remained relatively constant over the last 30 years (6) with increases reported in elite soccer (7). Moreover, reinjury rates are high (8,9) and thus injury prevention is paramount. However, knowledge about injury mechanisms is still limited. A better understanding of the movement conditions that pose a high risk for injury could improve efforts to prevent injury.

One possible injury mechanism is related to conditions under which the muscle is stretched to long lengths while under load. For example, the peak length of the hamstring muscle increases with running speed both at the muscle fiber (10) and muscle-tendon unit (11) levels. High running speeds also present large hamstring lengthening velocities under high loads (12). These eccentric loads are associated with increased muscle fiber strains (13) and eccentric work, predictors of microdamage (14–16). These biomechanical features of the hamstrings during high-speed running (large peak lengths, lengthening velocities, and eccentric loads) are thought to contribute to injuries (12,17,18) and may help explain the occurrence of injuries during high-speed running (19–22).

The eccentric loading and injury incidence associated with running has motivated the investigation of hamstring mechanics during high-speed running (23). For example, kinematic analyses have shown that hamstring lengths (normalized by the length in an upright posture) peak in late swing phase and are greatest in the biceps femoris long head compared to the other hamstring muscles due to moment arm differences (11,24). Kinetic analyses have demonstrated that hamstring loads may exceed 2000 N (25) and are greater in swing phase (10.5-26.4 N/kg) than in stance phase (4.6-11.6 N/kg) (23). Moreover, eccentric work is done exclusively in swing

phase (24,26) and hamstring muscle activity during stance phase is greatest at foot contact (27).

Previous investigations of hamstring injury mechanisms have primarily considered running at relatively constant and high speeds (23,28). However, hamstring injuries often occur when athletes are accelerating (29) and not necessarily at high running speeds (22). Moreover, the increase in injury incidence in elite soccer players between 2001-2014 (30) may partly result from the observed increase in the number of high-acceleration sprints performed by soccer players within a similar time frame (2006-2013) (31).

While the biomechanics of accelerative running have been explored (32–35), the kinematics of the hamstrings (muscle-tendon length and lengthening velocity) during acceleration are less understood. The hamstrings are important for generating propulsive ground reaction forces during acceleration (33,34), with large hamstring muscle forces (6-8 bodyweights) during a sprint that starts from stationary (33). Schuermans et al. (2017) identified increased anterior pelvis tilt and frontal plane trunk bending during the acceleration phase of sprinting as potential injury risk factors (36). Higashihara et al. (2018) found the biceps femoris was more active than the semitendinosus during stance phase in accelerative running, whereas the semitendinosus was more active in swing phase in both accelerative and top speed running (37). However, this study considered accelerative running at 15 m after the start of the sprint wherein athletes had already obtained a relatively high speed (an average of 8.54 m/s, approximately 90% of the average top speed of 9.52 m/s). Schuermans et al. (2017) observed a similar phase of the sprint (15-25 m after the start), but running speeds were not reported. Thus, the effect of running speed on hamstring mechanics (i.e., both kinematics and kinetics) during accelerative running remains unclear. Furthermore, the mechanics of a runner accelerating from a relatively high speed may not closely represent accelerative running activity during gameplay,

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which often is initiated from submaximal running speeds (a “flying” sprint). Finally, hamstring lengths and lengthening velocities during accelerative running have not been explored using a 3-dimensional musculoskeletal model similar to the work of Thelen et al. (2005) for non-accelerative, high-speed running on a treadmill (11).

Given the lack of knowledge about hamstring lengths and lengthening velocities during accelerative running, we aimed to compare these variables during accelerative running to running at a constant speed across a range of speeds. To emulate in-game running activity, we used a markerless motion capture technology (OpenCap (38)) that allowed analysis of overground accelerative running in an unconstrained outdoor setting. We hypothesized the hamstrings would be subject to greater peak muscle-tendon lengths and lengthening velocities in accelerative running compared to constant-speed running.

METHODS

Data Collection

Ten participants volunteered to participate in this study (Table 1). They were recreationally active with an average self-reported exercise frequency of 5 days per week for 1 hour per day. Following a self-directed warm-up, participants performed a 30-meter maximal effort sprint, 6 constant-speed running trials, each at a different speed, and 6 accelerative running trials. The 30-meter maximal effort sprint was used to determine the subject-specific top running speed. The accelerative running trials were flying sprints wherein participants began running at a constant speed and, when given a verbal cue, accelerated forward as fast as possible. These trials were designed to represent in-game accelerative running activity in team sports where sprinting is often initiated from a moving start. The first two running speeds were self-selected (jogging

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and brisk running) and the last 4 were prescribed (3, 4, 5, and 6 m/s, performed in that order). At each speed, subjects performed the constant-speed running trial first followed by the accelerative running trial and were given as much rest time between trials as desired to avoid fatigue. For each accelerative running trial, participants matched the constant speed in the previous trial but were then encouraged to accelerate as fast as possible. The order of running speeds and conditions was not randomized. All trials were performed overground on an outdoor running surface. All study activities were approved by the Stanford University Institutional Review Board. Informed written consent was obtained from all study participants.

For the jogging self-selected speed, participants were instructed to run at a comfortable jogging speed. For the brisk-running self-selected speed, they were instructed to run at a brisk running speed that was faster than the jogging speed, but not a maximal effort sprint. Participants matched prescribed speeds by self-modulating their running speed such that they passed cones positioned along the running lane at uniform distances (2.5 or 5.0 meters) to the beat of a metronome. The metronome frequency was based on the distance between cones and the prescribed speed (e.g., 60 beats per second with 5-meter cone spacing for a 5 m/s speed prescription). Subjects practiced running at each speed before data was collected. It was not imperative that speeds were matched exactly, but that a range of speeds were captured to allow investigation of the effects of running speed.

Two mobile phones were used to record videos (60 frames per second) of the running trials. The phones were attached to tripods placed such that the center of the capture volume was approximately 35 meters from the start of all running trials. Each phone was positioned approximately 5 meters from this 35-meter mark at an angle from the direction of travel on each side of the running lane ensuring both cameras could view the anterior-frontal plane along with

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the right-sagittal plane for one and the left-sagittal plane for the other. A diagram of the experimental setup is available in Supplemental Figure 1 (Supplemental Digital Content, http://links.lww.com/MSS/D104).

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Data Analysis

Video recording, synchronization, and processing as well as the scaling and kinematic analysis of a 3-dimensional musculoskeletal model (39) were performed with OpenCap (38) using OpenPose (40) as the underlying pose estimation model. We reprocessed the video data to leverage OpenPose's high accuracy settings, which were shown to provide more accurate results than the default settings at the expense of computational cost (38). The musculoskeletal model was scaled using video data from a static standing calibration trial that determined the lengths of the body segments. We identified consecutive foot contact events to segment kinematic data into individual steps. Foot contact events were identified as the instant coincident with the negative-going zero crossing in the shank sagittal plane angular velocity (lowpass filtered with a 2 Hz cutoff frequency for event detection only). This event follows a relatively large-magnitude, positive signal associated with the swing phase (41) and thus is reliably identified algorithmically and facilitates a consistent step segmentation. We analyzed the 2 steps (1 full gait cycle) immediately following the initiation of acceleration in the accelerative running trials and the 3 steps closest to the center of the capture volume in the constant-speed trials.

The length of the biceps femoris (long head) muscle-tendon unit (MTU) was calculated using OpenSim (42) and the subject-specific musculoskeletal model. The MTU length is dependent only on the hip and knee joint angles, which were calculated within OpenCap and were lowpass filtered with a zero-phase shift, 4th order Butterworth filter with 15 Hz cutoff

frequency adjusted for the backward pass (43). The 15 Hz cutoff frequency was chosen because, on average, 90% of the hip and knee flexion angle signal power was contained below 15 Hz across all running trials. We calculated running speed for each step by dividing the pelvis displacement during the step along the running direction (after lowpass filtering the pelvis translation signal with a 2 Hz cutoff frequency) by the step time. Hip and knee flexion speed were computed by numerically differentiating the hip and knee flexion angles with respect to time using the 5-point central difference method. The MTU lengthening velocity was calculated in OpenSim using the hip and knee joint angles and the hip and knee joint angle speeds as per

$$\sum_{i=1}^{4} r_i(t)\dot{\theta}_i(t) \#(1)$$

where $r_i(t)$ is the moment arm of the biceps femoris MTU about degree of freedom i at time instant t (dependent on hip and knee joint angles), $\dot{\theta}_i(t)$ is the time-derivative of the angular coordinate associated with degree of freedom i at time instant t, and the subscript i indexes the four relevant degrees of freedom (hip flexion, hip internal rotation, hip adduction, and knee flexion).

For each step from each trial, we used the peak MTU length and average lengthening velocity as primary outcomes to compare running conditions. We analyzed only the leg (right or left) that was exclusively in the swing phase during each step. For example, we considered the left leg for a step beginning with a right foot contact and ending with a left foot contact. In this case, the average MTU lengthening velocity would be calculated as the sample average of all MTU velocity data across the entire step for which the MTU of the left leg was lengthening (a positive MTU velocity). We chose to explore the average lengthening velocity instead of the peak velocity because it is more robust to noise introduced from numerical differentiation.

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Nonetheless, we also explored the peak lengthening velocity in a post-hoc analysis (results reported in the Supplemental Digital Content, http://links.lww.com/MSS/D104).

To account for size variations across participants, we normalized MTU lengths and lengthening velocities by the MTU length in the neutral configuration (hereon referred to as the neutral length). The neutral configuration is an upright standing pose. To account for variability in top running speed across participants, we normalized running speeds by the subject-specific maximum observed in the 30-meter sprint. We also calculated the following variables to explore the relationship between joint kinematics and the primary outcomes: hip flexion, knee flexion, pelvis tilt, and thigh tilt angles at the instant of peak MTU length and the contributions from the hip and knee flexion speed to the average MTU lengthening velocity. Average MTU lengthening velocity was determined by first computing the MTU velocity as in equation (1). The sample average was then computed across all MTU velocities for which the MTU was lengthening. We calculated the contribution of hip and knee flexion speed as the sample average of the coordinate-relevant products from equation (1) (i.e., for i = hip flexion and i = knee flexion) across all MTU velocities for which the MTU was lengthening. Pelvis and thigh tilt angles were calculated using an Euler angle decomposition (ZXY for the pelvis, ZYX for the thigh) of the rotation matrix representing the orientation of each segment relative to the global coordinate system. The tilt angle for both segments was the angle associated with the first rotation (about the Z-axis). Because the motion is primarily sagittal, the tilt angle approximates the angular deviation of the segment long axis relative to the global vertical axis. For example, the pelvis and thigh tilt angle would both be 0° during a standing posture, 90° during a supine posture, and -90° during a prone posture.

Data Quality Assessment

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We identified and removed 61 steps (out of 300 total) based on several data quality checks leaving 239 available for analysis (140 steps for constant-speed running and 99 steps for accelerative running). At least 1 step was removed for each participant. The maximum number of steps removed for a participant was 12. As the first step in the quality control process, we removed trials where the kinematic reconstruction failed, as evident following visual inspection in the OpenSim graphical user interface (2 constant-speed trials removed). Next, trials were removed if any foot contacts were not identified (4 accelerative trials and 3 constant-speed trials removed). Then, an individual step was removed if either outcome measure (peak length or lengthening velocity) for the step was deemed an outlier, called a discrete outlier, or if one of several other relevant kinematic time-series for the step was deemed an outlier, called a time-series outlier. A discrete outlier was defined as a value that was greater than 3 standard deviations away from the mean. For detection of time-series outliers, all time-series were first resampled to express their values as a function of the percentage of the step duration. A time-series was labeled an outlier if any of these discrete-time values met the discrete outlier criterion. The time-series included for outlier detection were the hip and knee flexion angle and speed and the MTU length and lengthening velocity. The mean and standard deviation used for outlier detection were computed using data from all participants, but specific to running condition (constant-speed or accelerative) and running speed (0-55%, 55-65%, 65-75%, 75-85%, and 85-100% of top speed).

Statistical Analysis

We used 2 separate linear mixed model regression analyses to compare the primary outcomes (MTU peak lengths and MTU lengthening velocities) between running conditions with running speed as the independent variable. Each regression model had 4 fixed-effects

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coefficients including 2 condition-specific intercepts and 2 condition-specific slopes (modeling the effect of running speed) along with 4 random effects (condition-specific slopes and intercepts) per participant. Running speeds were shifted (normalized running speed minus 1) for model fitting such that the intercept corresponded to the response (peak length or lengthening velocity) at top speed. We used the fitlme function in Matlab (R2021a, The MathWorks, Inc., Natick, MA) for model fitting using restricted maximum likelihood estimation (REML) and estimated all elements of the covariance matrix (i.e., a FullCholesky covariance pattern). To explore the effects of different joint and segment kinematics on the primary outcomes, we also fit 6 additional linear mixed regression models with the same structure as for the primary outcomes, but with different response variables: hip flexion, knee flexion, pelvis tilt, and thigh tilt angles at the instant of peak MTU length and the contributions from the hip and knee flexion speed to the MTU lengthening velocity.

RESULTS

The general pattern of the biceps femoris MTU length throughout the step was similar for constant-speed and accelerative running (Figure 1). Minimum MTU length occurred at approximately 25% of the step cycle. The hamstrings then lengthened up to a peak length at approximately 75-80% of the step cycle and terminated with a brief shortening phase just before foot contact (Figure 1, top row).

Despite similarity in the general pattern of the MTU length trajectories, accelerative running resulted in greater peak MTU lengths, with more pronounced differences for slower running speeds (Figure 1). At top speed, the peak lengths were 111-112% of neutral length (11-12% strain), and this was not different between conditions (p=0.431, Figure 2A). However, the

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peak MTU length was greater ($p<0.001$, Figure 2A) when accelerating from slow speeds than in speed-matched, constant-speed running (i.e., below 75% of top speed based on 90% prediction intervals in Figure 2A). For example, the regression analysis predicted that the peak MTU length experienced when a person accelerates while running at only 50% of top speed is the same as if running with a constant speed of 88% of top speed. (The linear mixed regression results are also tabulated in Supplemental Table 1, Supplemental Digital Content, http://links.lww.com/MSS/D104.)

The greater peak MTU lengths in accelerative running resulted from a more flexed hip (hip flexion lengthens the hamstrings), despite a more flexed knee (knee flexion shortens the hamstrings) compared to constant-speed running (Figure 3). The MTU lengthening effect of hip flexion was greater than the shortening effect of knee flexion because the hamstring moment arm about the hip is greater than the knee. The differences between accelerative and constant-speed running in the hip and knee flexion angle at the instant of peak MTU length were greater for slower running speeds (Figure 1, middle and bottom rows). For example, the hip was more flexed ($p<0.001$) in accelerative than in constant-speed running at speeds below 96% of top speed (based on overlap of the 90% prediction intervals, Figure 3A). This greater hip flexion angle at the instant of peak MTU length was due primarily to a more horizontal thigh (Figure 4B) in addition to a more anteriorly tilted pelvis (Figure 4A). Based on the linear mixed model, the hip was flexed 6° more in accelerative running than in constant-speed running with every 10% decrease in running speed, where 4° of the difference was due to a more horizontal thigh and 2° to a more anteriorly tilted pelvis.

The average MTU lengthening velocity was greater during accelerative than in constant-speed running across all running speeds (Figure 2B). This resulted from a greater hip flexion

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speed, rather than from a greater knee extension speed (Figure 5). In fact, below 78% of top running speed, the MTU lengthening velocity resulting from the knee extension speed was greater in constant-speed than in accelerative running (based on 90% prediction interval overlap, Figure 5B).

DISCUSSION

Our results demonstrate that the hamstrings are stretched more and faster during accelerative compared to constant-speed running. Furthermore, the differences were more pronounced when accelerating from slower running speeds. This provides a potential biomechanical perspective for understanding the occurrence of hamstring injuries during accelerative running (29) (and not necessarily at high running speeds (22)). However, further research is necessary to establish causal relationships between acceleration biomechanics and hamstring injury. Our results also have implications for athlete monitoring which often use high-speed running as an index of load (44,45). For example, while our data support the use of exposure to high running speeds (e.g., from body-worn GPS units) as a proxy for increased hamstring length and lengthening velocity, they also suggest the accelerative nature of running should be included to more comprehensively characterize exposure to high-risk circumstances.

Our results illustrate how hip-knee coordination underlies the differences between accelerative running and constant-speed running and the implications for exposing the hamstrings to long lengths. The kinematics of accelerative running were such that the peak MTU length occurred at an average hip flexion angle of 67° (Figure 3A), and this was not influenced by running speed. However, the knee flexion angle at which the peak MTU length occurred was influenced by running speed. For example, when accelerating from slower running speeds, the

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knee was more flexed at the instant of peak MTU length than in speed-matched, constant-speed running (approximately 3° more knee flexion with every 10% of top speed decrease in running speed). Without this increase in knee flexion, the peak MTU length would have been even greater when accelerating from slower running speeds (i.e., when the capacity to accelerate is greater). Thus, a coordinated motion wherein the hip and knee flex together during the late swing phase of accelerative running reduces the peak length of the hamstrings. In contrast, an accelerative running pattern wherein the knee does not flex synchronously with the hip exposes the hamstrings to longer peak lengths.

The importance of hip-knee coordination in accelerative running is further supported by the observed MTU lengthening velocities. Large hip flexion speeds drove the increase in MTU lengthening velocity during accelerative running compared to constant-speed running. This was despite the knee extension speed contributing less to MTU lengthening velocity in accelerative compared to constant-speed running (Figure 5B). In fact, our linear mixed models predicted that when accelerating from below approximately 40% of top speed, the knee contributed a net average *shortening* velocity (negative vertical axis in Figure 5B). Therefore, a lack of simultaneous hip and knee flexion during accelerative running would be two-fold detrimental. It would result in greater peak MTU length and greater MTU lengthening velocity. An altered hip-knee coordination in this way is generally consistent with others who have noted the potential influence of altered coordination on injury risk (28).

The strong influence of hip joint kinematics on peak MTU length we observed was also reported recently by Astrella et al. (2024) (46). Their data predicted a peak biceps femoris MTU strain of 8.9% at the greatest acceleration level considered (the first step of a 40-meter sprint at which the average running speed was approximately 40% of top speed); this compared well with

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our data (9% at 40% of top speed during accelerative running). However, while we found peak MTU strain slightly increased when accelerating from faster speeds, their data predicted a slight decrease with increasing running speed. This may be explained by differences in the accelerative tasks between the two studies. We considered a flying sprint during which acceleration was preceded by constant-speed running with submaximal effort, whereas the steps analyzed in Astrella et al. (2024) were preceded by maximal effort accelerative running.

While our focus was on kinematic variables, future work should also compare muscle kinetics between accelerative running and constant-speed running. Cumulative eccentric MTU work has been proposed as a predictor of muscle damage and would be a candidate proxy for hamstring injury risk. Our use of average MTU lengthening velocity may be considered a kinematic surrogate for eccentric work since eccentric work is proportional to the average MTU lengthening velocity weighted by MTU force. Therefore, if hamstrings activation and loading exhibit a similar pattern between the two tasks, one would expect similar differences in eccentric work as for average lengthening velocity. Previous investigations of constant-speed running suggest hamstrings are relatively inactive at the beginning of the lengthening phase and become more active towards the end (37,47,48). Although a similar pattern has been observed for accelerative running (37), further research is necessary to quantify differences in muscle eccentric work between running conditions.

OpenCap allowed us to measure the kinematics of accelerative running overground in outdoor settings. While this system has been validated for a variety of tasks, including slower-speed running, a limitation of our study is that OpenCap has not been validated for the specific tasks considered in our study. The ability of OpenCap and OpenSim to accurately estimate MTU length and lengthening velocity for these tasks, however, is supported by the similarity of our

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results in comparison with several previous studies. Thelen et al. (2005) reported peak MTU lengths between 1.09-1.10 neutral lengths, which occurred at hip and knee flexion angles between 63-65° and 44-45°, respectively, during constant-speed running at 80-100% of top speed (11). Under the same conditions, the linear mixed models based on our data predict similar results for peak MTU length (1.09-1.11 neutral lengths) and hip flexion (between 52-64°) but with less knee flexion (28-35°). We observed the same pattern of the MTU length trajectories (e.g., compare the top row of our Figure 1 with the top row of Figure 2 from Thelen et al. (2005)). Schache et al. (2012) report peak biceps femoris MTU strain of 12% at top speed (24) which is comparable to our results (11%). Schache et al. (2013) report a peak MTU lengthening velocity of 2.2 neutral lengths per second for sprinting at 9.0 m/s (47). Our linear mixed effects models predict the same value for running at top speed (Supplemental Figure 3, Supplemental Digital Content, http://links.lww.com/MSS/D104). Moreover, we observed the same general pattern of MTU lengthening velocity trajectories (e.g., compare the top row of Supplemental Figure 2, Supplemental Digital Content, http://links.lww.com/MSS/D104, with the right column of Figure 3 from Schache et al. (2013)). The linear models based on our data predict anterior pelvis tilt angles between 14-19° across both accelerative and constant-speed running at top speed, which is comparable to the range reported by previous studies (e.g., 10-20° (49) and 15-30° (36)). We report hip and knee angle trajectories comparable to Higashihara et al. (37) with slightly lower peak values which may arise because the average top speeds of the sprinters in their study (9.52 m/s) is higher than ours (7.4 m/s).

Our study is further limited by a small sample size and because not all athletes specialized in sprinting. Furthermore, we considered only the step cycle (as opposed to stride cycle) of accelerative running where running speed was increased (positive acceleration). The

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portion of the swing phase within the step cycle that we considered fully contains the lengthening action of the hamstrings (Figure 1) and thus is sufficient to capture the main kinematic variables in this study. However, there may be differences between conditions during stance phase with implications on injury risk that should be considered in future work. Furthermore, hamstring injuries have also been observed in decelerative running (22,29) and a recent analysis of joint kinetics identified some mechanical features of accelerative and decelerative running that help explain hamstring injury risk during those tasks (50). However, it remains unclear how hamstring mechanics differ between decelerative compared to accelerative or constant-speed running and should be examined in future work. Finally, we did not include sex as a factor in our statistical model, but it may be warranted in future research due to sex-dependent morphology that affects hamstring kinematics (e.g., pelvis shape).

CONCLUSIONS

The hamstrings are stretched to greater peak lengths and are subject to greater lengthening velocities in accelerative than in constant-speed running. Greater hip flexion and hip flexion speed during accelerative running was the primary driver of the greater hamstring lengths and velocities. Moreover, these effects were more pronounced when accelerating from slower running speeds. Thus, the accelerative nature of running should be considered in addition to running speed for monitoring athlete exposure to high-risk circumstances. These high-risk circumstances can be mitigated in part by coordinating the hip and knee motions during accelerative running.

Acknowledgments

The results of this study are presented clearly, honestly, and without fabrication, falsification, or

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inappropriate data manipulation. The results of the present study do not constitute endorsement by the American College of Sports Medicine. This work was supported by the Wu Tsai Human Performance Alliance at Stanford University and the Joe and Clara Tsai Foundation, and the National Institutes of Health through Grant P41EB027060. We thank Julie Muccini for her help running experiments.

Author Contributions

RG, ZT, JH, SD conceived the study and designed the experiments. RG and ZT collected data. RG, ZT, and AF analyzed data. All authors contributed to writing the manuscript.

Conflicts of Interest

AF is the cofounder of Model Health, Inc., which supports the non-academic, commercial use of the open-source software used for data collection in this study.

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FIGURE LEGENDS

Figure 1. Biceps femoris MTU length (top row), hip flexion angle (middle row), and knee flexion angle (bottom row) as a percentage of the step duration. The step duration is approximately one half of the running cycle, and the results are plotted for the leg that is exclusively in swing phase, as shown in the lower left panel. Results are compared for accelerative running (red) and constant-speed running (teal). Solid lines and shaded areas indicate the ensemble mean +/- standard deviation across all participants within the same speed bin (indicated at the top of each column). Running speeds are normalized by the subject-specific top speed.

Figure 2. Peak biceps femoris MTU length (A) and average lengthening velocity (B) vs. normalized running speed during both constant-speed (teal) and accelerative (red) running. Neutral length is the MTU length in the neutral configuration (upright standing posture). Each data point (triangles for constant-speed, circles for acceleration) represents the leg exclusively in swing phase during a single step. Solid lines depict the linear mixed model regression lines, and the shaded area illustrates the 90% prediction interval.

Figure 3. Hip (A) and knee (B) flexion angle at the instant of peak MTU length vs. normalized running speed during both constant-speed (teal) and accelerative (red) running. Each data point (triangles for constant-speed, circles for acceleration) represents the leg exclusively in swing phase during a single step. Solid lines depict the linear mixed model regression lines, and the shaded area illustrates the 90% prediction interval.

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Figure 4. Pelvis (A) and thigh (B) tilt angle at the instant of peak MTU length vs. normalized running speed during both constant-speed (teal) and accelerative (red) running. Positive pelvis and thigh tilt angles both contribute to hip flexion. For example, at top speed (normalized running speed = 1) in accelerative running, the pelvis (19°) and thigh (47°) tilt angles sum to 66°, the approximate hip flexion angle in the same condition (Figure 3A). Each data point (triangles for constant-speed, circles for acceleration) represents the leg exclusively in swing phase during a single step. Solid lines depict the linear mixed model regression lines, and the shaded area illustrates the 90% prediction interval.

Figure 5. Hip flexion (A) and knee extension (B) speed contributions to the average MTU lengthening velocity vs. normalized running speed during both constant-speed (teal) and accelerative (red) running. Neutral length is the MTU length in the neutral configuration (upright standing posture). Each data point (triangles for constant-speed, circles for acceleration) represents the leg exclusively in swing phase during a single step. Solid lines depict the linear mixed model regression lines, and the shaded area illustrates the 90% prediction interval.

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Figure 1

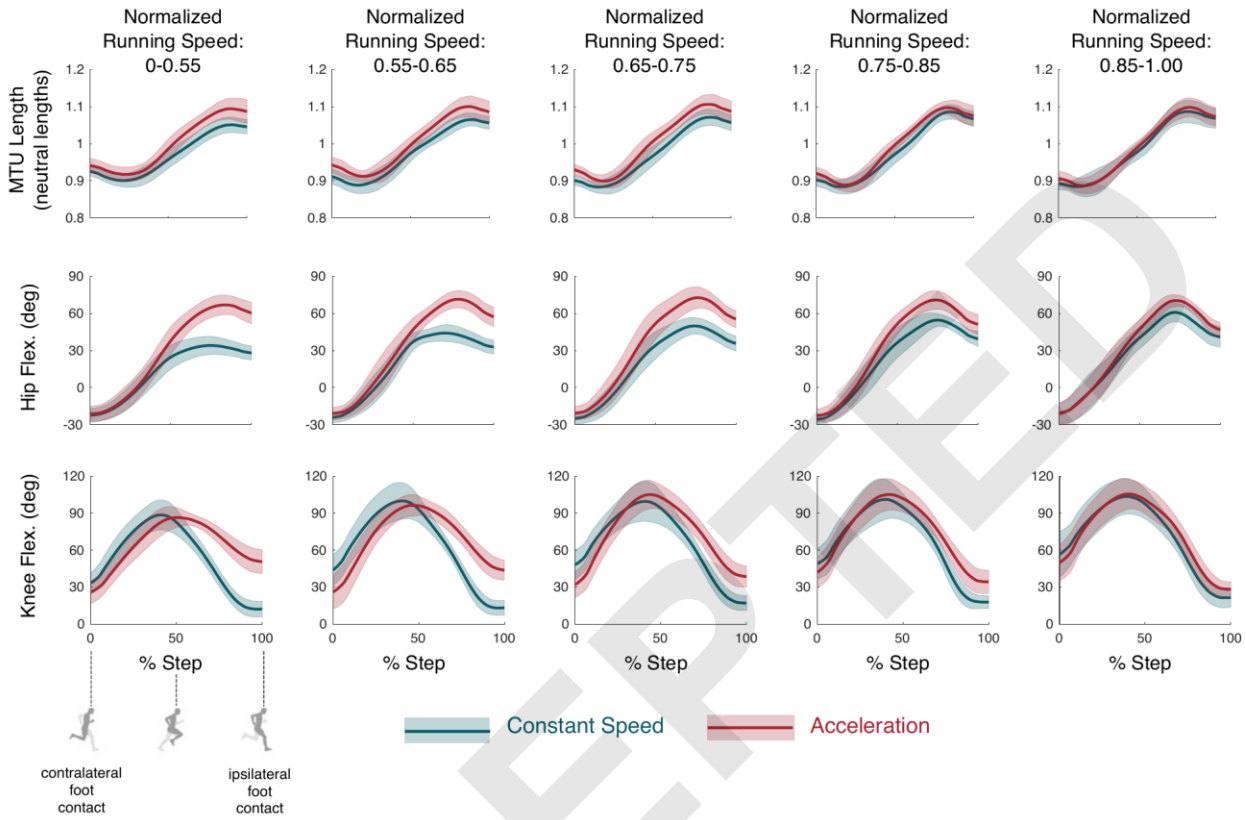

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Figure 2

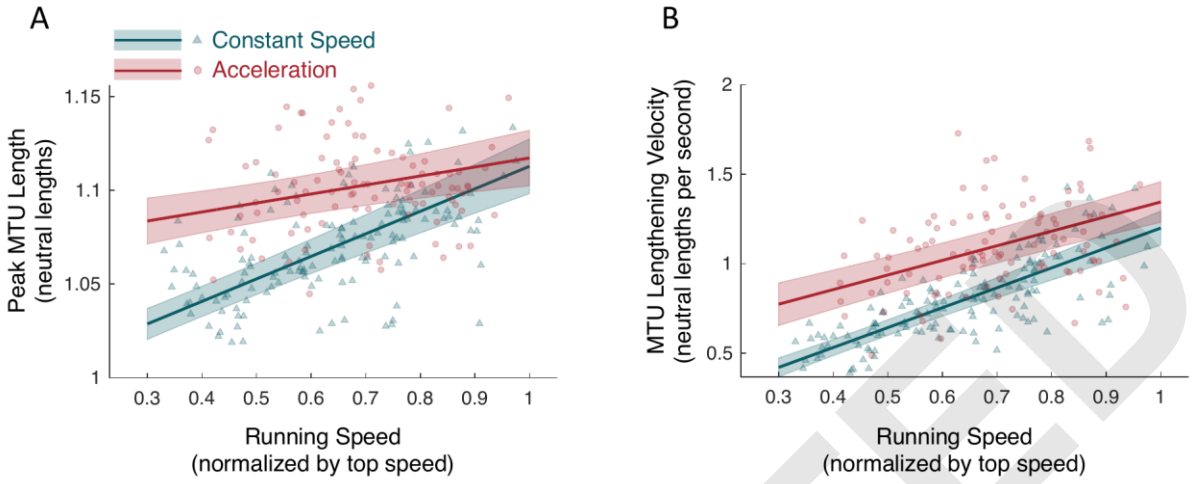

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Figure 3

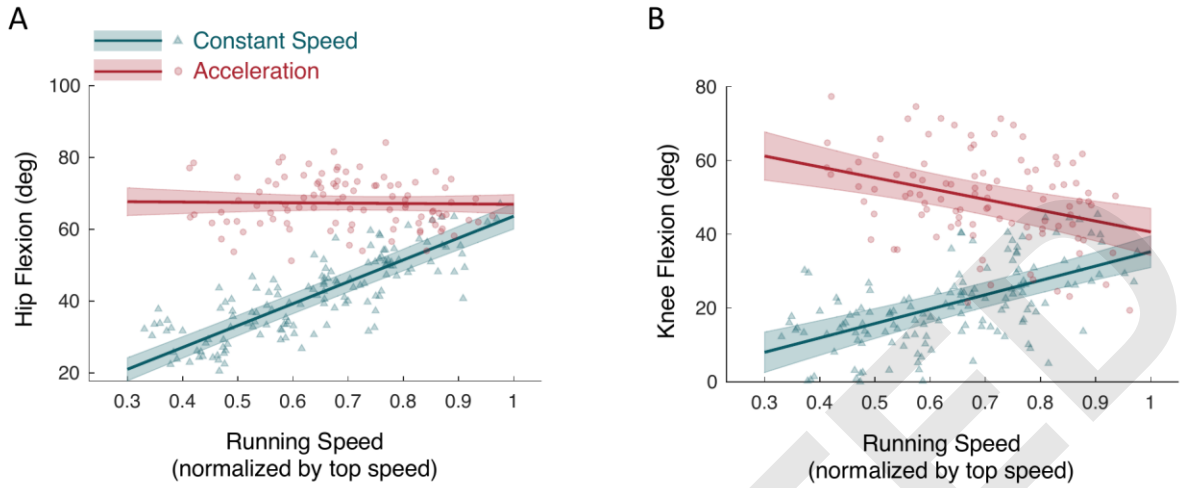

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Figure 4

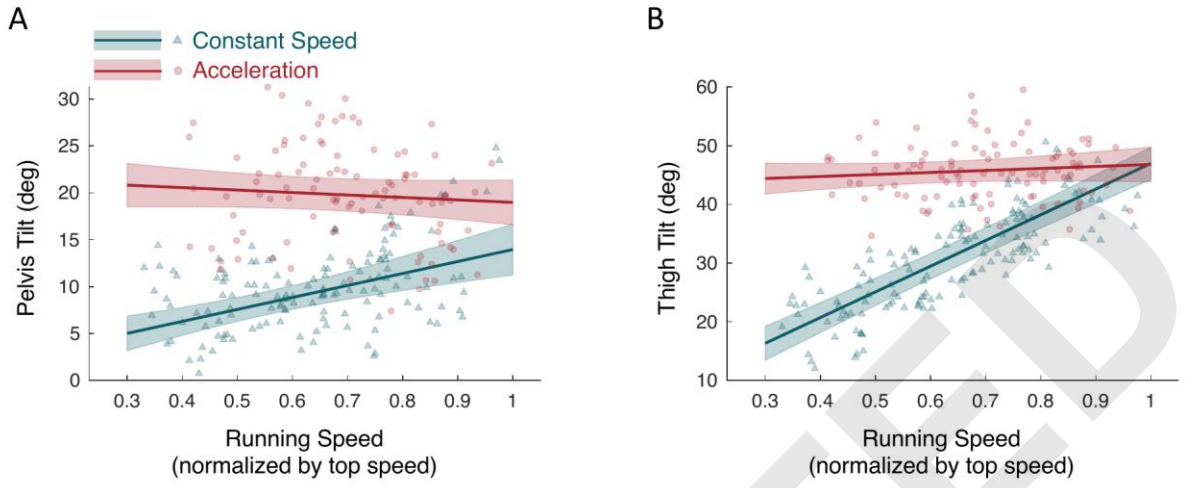

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Figure 5

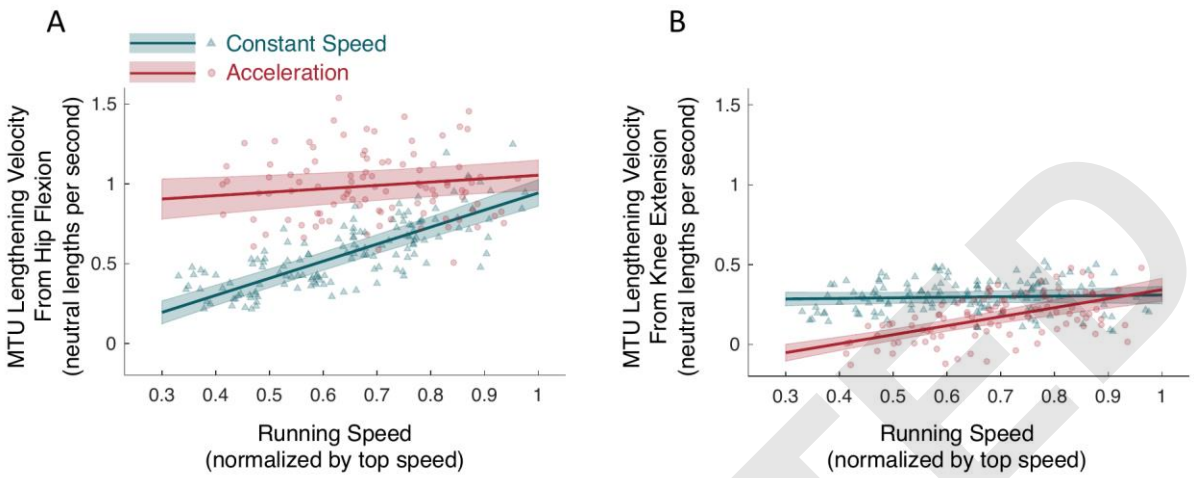

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Table 1

Subject characteristics

	Male Mean (SD)	Female Mean (SD)	All Sexes Mean (SD)
Sample Size	5	5	10
Age (years)	23.3 (4.5)	21.8 (0.9)	22.6 (3.2)
Height (cm)	177.8 (8.6)	168.7 (8.2)	173.2 (9.3)
Mass (kg)	79.4 (16.0)	66.7 (8.7)	73.0 (13.9)
Top Speed* (m/s)	7.9 (0.5)	6.9 (0.5)	7.4 (0.7)

*As observed in a maximal effort 30-meter sprint

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Supplementary Material

Table S1. Linear mixed model fixed effects coefficients. Significant (p<0.05) differences are bolded. Intercept units for peak MTU length are in neutral lengths, for MTU lengthening velocity and MTU lengthening velocity from knee and hip are in neutral lengths per second, and for hip and knee flexion and pelvis and thigh tilt angle at peak MTU length are in degrees. Slope units are the intercept units per normalized running speed. As noted in the main text, running speeds were shifted for model fitting such that the intercept corresponded to the response at top speed. Thus, for example, the model for peak MTU length during constant speed running is y = 1.11 + 0.12(x-1).

	Intercept			Slope		
Response	Constant	Accel.	Difference	Constant	Accel.	Difference
Peak MTU Length	1.11	1.12	0.00 (p=0.431)	0.12	0.05	**-0.07 (p<0.001)**
MTU Lengthening Velocity	1.20	1.35	**0.15 (p=0.012)**	1.11	0.82	**-0.30 (p=0.045)**
Hip Flex. at Peak Length	63.65	66.98	3.33 (p=0.079)	60.90	-1.02	**-61.92 (p<0.001)**
Pelvis Tilt at Peak Length	13.96	18.98	**5.03 (p=0.003)**	12.77	-2.63	**-15.40 (p<0.001)**
Thigh Tilt at Peak Length	46.98	46.80	-0.19 (p=0.906)	43.80	3.42	**-40.37 (p<0.001)**
Knee Flex. at Peak Length	35.26	40.62	5.36 (p=0.088)	38.95	-29.37	**-68.32 (p<0.001)**
MTU Velocity from Knee	0.31	0.34	0.03 (p=0.291)	0.04	0.56	**0.53 (p<0.001)**
MTU Velocity from Hip	0.94	1.05	**0.11 (p=0.023)**	1.07	0.21	**-0.86 (p<0.001)**

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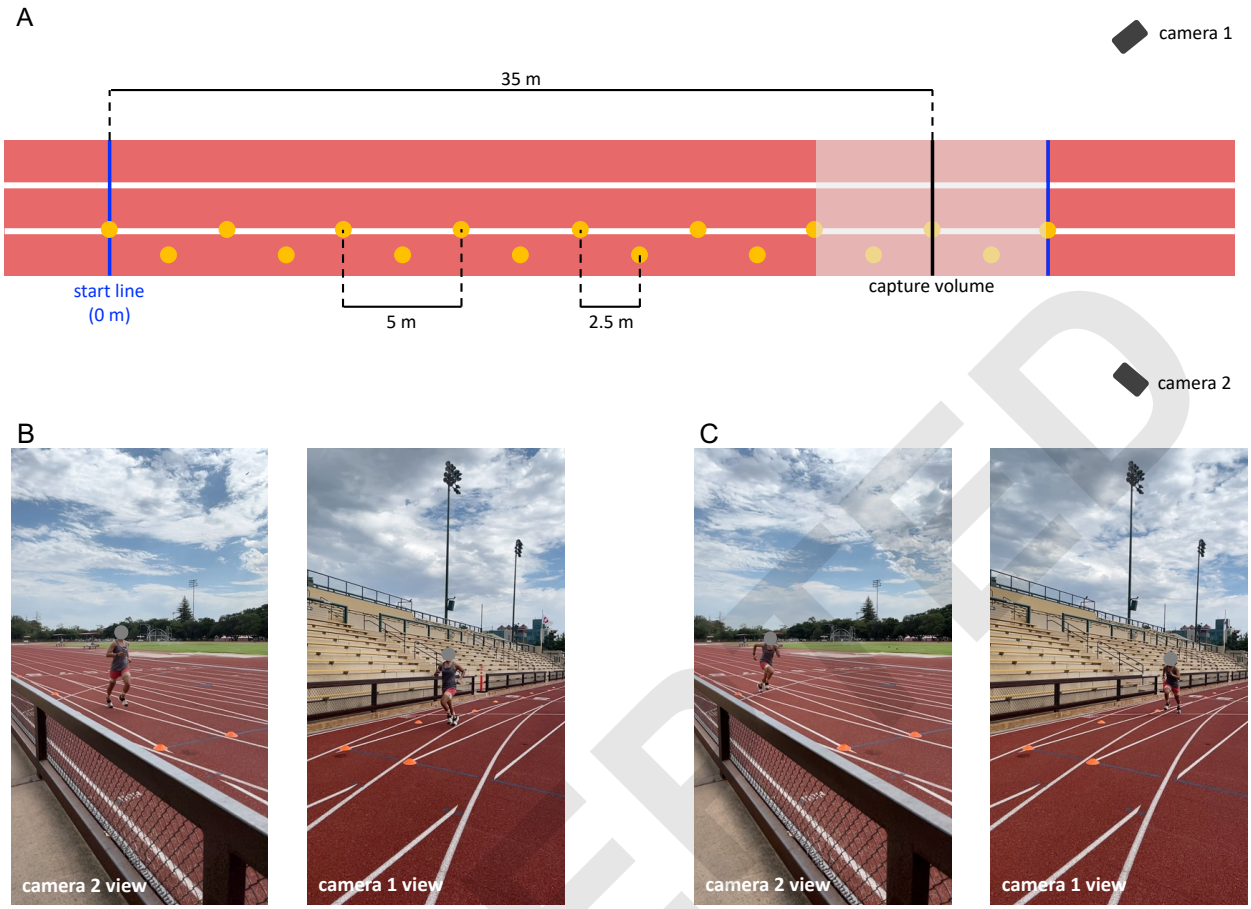

Figure S1. Experimental setup (A). Orange dots represent cones placed along the side of the running lane with 5 and 2.5 meter spacing as depicted. The field of view of each camera is illustrated for an example participant during a constant speed running trial in panel B and an accelerative running trial in panel C.

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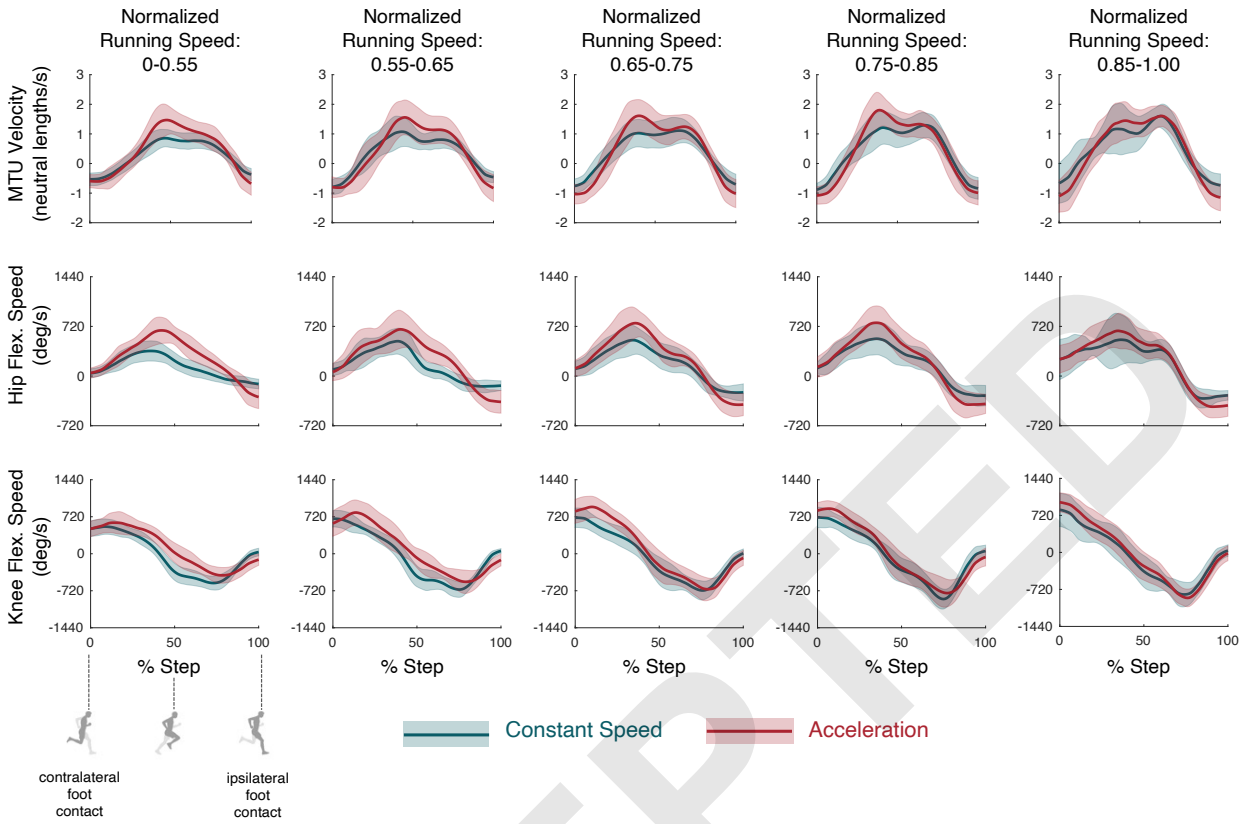

Figure S2. Biceps femoris MTU velocity (top row, units: neutral lengths per second), hip flexion speed (middle row, units: degrees per second), and knee flexion speed (bottom row, units: degrees per second) as a percentage of the step duration for the accelerative (red) and constant speed (teal) running conditions. Solid lines and shaded areas indicate the ensemble mean +/- standard deviation across all participants within the same speed bin (indicated at the top of each column). Running speeds are normalized by the subject-specific top speed. Positive MTU velocities are lengthening velocities (negative are shortening velocities). Positive hip and knee flexion speeds are flexion speeds (negative values are extension speeds).

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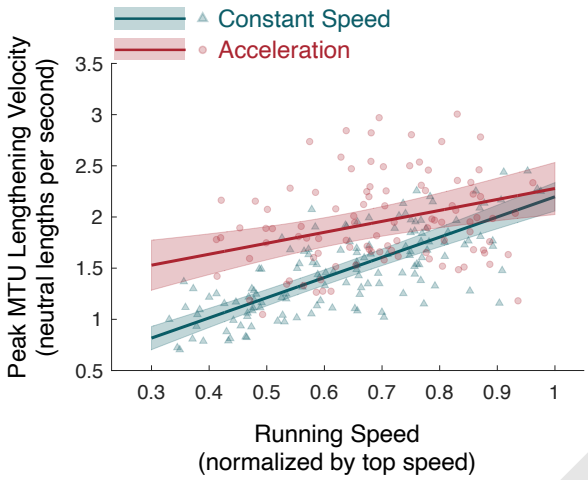

Figure S3. Peak biceps femoris MTU velocity vs. normalized running speed during both constant speed (teal) and accelerative (red) running. Neutral length is the MTU length in neutral configuration (upright standing posture). Each data point (triangles for constant speed, circles for acceleration) represents a single step. Solid lines depict the linear mixed model regression lines, and the shaded area illustrates the 90% prediction interval. The slope and intercept of the linear mixed models for both accelerative and constant speed running are shown in the figure. While the peak lengthening velocity at top speed (intercept) was not significantly different (p=0.54) between conditions (2.20 and 2.28 neutral lengths per second for constant speed and accelerative running, respectively), the slope of the accelerative model (1.07 neutral lengths per second per normalized running speed) was significantly less (p<0.01) than the constant speed model (1.97 neutral lengths per second per normalized running speed).

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